## Sundararajan Narasiman

Principal Enterprise Architect, Ex- Amazon, Ex-AWS
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## Professional Summary

Results driven hands-on Principal Software Architect with 20+ years of experience in designing, developing and optimizing large scale software platforms. Proven track record of driving technology innovation, improving code quality and dev process hygiene and mentoring architects/engineers across org. Passionate about continuous learning and challenging the status quo.

Expertise in Cloud Native Architecture, Cloud Platform Engineering, App Modernization, Microservices, Micro Frontend, MLOps.

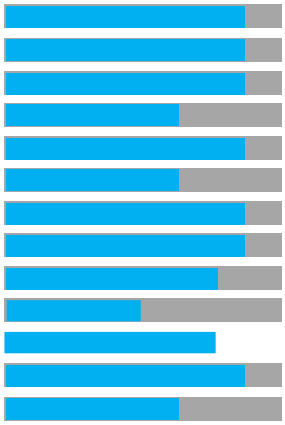
Proven recording of performing both breadth-role and depth-role in architecture/technology and demonstrating results. Passionate about nurturing global technology communities and recipient of Microsoft Most Valuable professional for 7 years (2008-2014).

Demonstrated thought leadership through AWS public speaking, AWS open-source contribution and AWS blogs.

Proven record of incubating and building from zero to one.

## Core Competencies

- Enterprise Architecture
- System Design
- Cloud Platform Architecture
- Database Design
- Microservices Architecture Patterns
- Data Engineering
- ML Ops & DevOps
- Mentoring & Coaching
- Team Building & Org development
- AI/ML
- API Design & Development
- Site Reliability Engineering
- Data Structures & Algo



## Professional Experience

**Principal Enterprise Architect/ Vice President**
**Computer Age Management Services | Chennai |August 2024 to till date**

- Architected and built a GenAI based knowledge comprehension tool to solve the problems of 19+ AMCs across India. Solution relied on RAG based knowledge retrieval from Google Gemini LLM

- Improved developer productivity by integrating hosted OLAMA LLM models into IDEs. This led to cost savings of 80% compared to licensed code assist tools
- Performed periodic code reviews and ensured all the critical NFR gaps are addressed in the code before UAT
- Architected and built strategic Cloud Foundation on GCP for futuristic Mutual Fund RTA platform for India market in 5 months, complying with SEBI regulations
- Practiced Domain Driven Design by de-composing a legacy monolith to 30+ Microservices for better manageability and scale
- Conceptualized and led API rationalization exercise to optimize core APIs from 130 to 75 and completed design of the same
- Unblocked engineering teams by building re-usable components (11), framework libraries (9), platform services (15)
- Hired and built a team of 6 Staff Architects across app, data, ai and infrastructure
- Mentored and coached staff architects and staff engineers

**Principal System Dev Engineer**
**Amazon Development Center | Chennai/Bangalore |October 2022 to June 2024**

- Architected and built a container orchestration platform from ground up for a media streaming use case
- Identified security gaps in streaming architecture, collaborated with AWS Shield team and built a custom feature to harden platform
- Operationalized a Tier-1 service, identified AWS Well Architected gaps, implemented SRE practices and demonstrated platform availability improvements from 75 to 99.9
- Identified performance bottlenecks, built a custom scheduling algorithm and improved container density from 5 to 25, which reduced EC2 fleet by 3X leading to infrastructure cost savings by more than 50%
- Built a custom container scheduling algorithm for media streaming use case and filed an innovation disclosure
- Seeded Automate Everything mindset, implemented IaCs to reduce platform deployment time to 30 minutes from 8 hours
- Re-designed and migrated streaming platform code to support ARM based architecture (AWS Graviton). It resulted in infrastructure cost savings by more than 70%
- Mentored and coached SysDevs and SDEs

**Principal Solutions Architect**
**Amazon Web Services | Chennai/Bangalore |March 2017 to September 2022**

- Architected and implemented automated MLOps using AWS SageMaker for a leading US based manufacturing organization and reduced cycle time for productionizing ML Models from days to hours
- Re-engineered a legacy partner API for a FinTech and re-designed using AWS Cloud Native Architecture and Microservices. Demonstrated scalability improvements from 300 TPS to 1000 TPS
- Modernized contact center platform for leading US based insurers using Amazon Connect based stack. Demonstrated call volume improvements by 20% and increase in CSAT by 25%
- Re-architected and SAASified a legacy MF platform for a FinTech using AWS SAAS Factory approach. Increased customer acquisition by 30%, aiding in business transformation goals.
- Designed and built a cloud-based video monitoring solution for home automation giants by leveraging AWS IoT stack which reduced infrastructure costs by 25%
- Architected and led API rationalization effort for a leading Fintech and consolidated APIs from 1300+ to 350+ which reduced operational overhead by 35%

- Re-engineered and modernized a legacy food ordering platform by leveraging AWS Serverless architecture and demonstrated infrastructure cost savings by 90%. Improved efficiency by reducing unit transaction cost charged to end customers by 40% and shortened cycle time for new region addition to the platform from months to weeks.

**Sr. Advanced Systems Engineer**
**Honeywell Technology Solutions | Madurai/Bangalore |Oct 2014 to Feb 2017**

- Modernized a legacy home automation product in MS Azure and supported a achieved a TPS of 1000
- Transformed home automation product into a global platform and scaled to 3 markets (US, India and Europe) with a CAGR of 20%
- Designed and built first version of Honeywell IOT Device Management platform on MS Azure which supported 5000 devices in 1st year
- Achieved NFR coverage of > 90% by ensuring 'ility' scorecards are followed & tracked throughout NPI life cycle
- Improved security posture of a legacy platform by ensuring zero sev1 and sev2 cyber sec vulnerabilities

**Sr. Architect - Technology**
**Cognizant Tech Solutions | Chennai |Aug 2006 to Sep 2014**

- Incubated MS Azure, built accelerators and delivered mission critical projects
- Architected and built couple of mission critical projects for leading insurers using Enterprise Architecture best practices

**IT Analyst**
**Tata Consultancy Services | Chennai |Feb 2001 to July 2005**

- Performed functional enhancements of core underwriting engine of leading life insurers in USA
- Bug fixes of core mainframe COBOL modules and unit testing

## Technical Skills

Programming Languages: Java, Python, C#, NodeJs , GoLang
Frameworks & Libraries: React, .NET core. ASP.NET Core, ASP.NET Web API, PyTorch, ScikitLearn, AWS SageMaker
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD
Databases: MySQL, Redis, AWS DynamoDB, AWS Document DB, AWS KeySpaces
AI/ML: Deep Learning, NLP, Reinforcement Learning, Computer Vision, FlanT5, Prompt Engg
Architectural Patterns: Microservices, Event-Driven Architecture, Serverless, SOA
Data Analysis Tools: Pandas, NumPy, MatPlotLib
CNCF Stack: Isio, Prometheus, Graffana
AWS Services: App Mesh, RDS, Transcribe, Polly, Lex, Translate, Glue, Kinesis, API Gateway, StepFunctions, Lambda
GCP Services: GKE, Functions, Gemini, Apigee

## Soft Skills

Technical Leadership, Cross-functional collaboration, Mentoring, Agile, Effective Communication, Adaptability, Innovation, Organizational Development, Adaptability to new Technologies

## Education

- Professional Certificate in Product Management from SP Jain Institute of Management (Nov 2023 – May 2024)
- Bachelor of Engineering in Instrumentation & Control (1996-2000)
  from Madurai Kamaraj University with 70% CGPA
- H.S.C from Tamilnadu Education Bord (1994-1995) with 87% CGPA
- S.S.L.C from Tamilnadu Education Bord (1992-1993) with 77% CGPA

## Certifications

- AWS Certified Developer Associate
- AWS Certified Solution Architect Associate
- AWS Certified SysOps Associate
- AWS Certified DevOps Engineer – Professional
- AWS Certified Solution Architect – Professional
- Microsoft Certified Azure Developer
- Certified Scrum Master
- Microsoft Certified Application Developer (MCAD)

## Awards

- Awarded by Microsoft as the **Most Valuable Professional** in Microsoft SharePoint for 7 years from 2008 to 2014 for outstanding contributions to Global Technology Community
- Awarded by Microsoft as **Microsoft Community Contributor** in 2014
- Awarded by DZone as **Most Valuable Blogger** since 2014

## Thought Leadership - Blogs

Published the following blogs in AWS channel as primary author.
https://aws.amazon.com/blogs/devops/automated-ci-cd-pipeline-for-net-core-lambda-functions-using-aws-extensions-for-dotnet-cli/
https://aws.amazon.com/blogs/compute/migrating-net-classic-applications-to-amazon-ecs-using-windows-containers/
https://aws.amazon.com/blogs/compute/hosting-asp-net-core-applications-in-amazon-ecs-using-aws-fargate/

## Projects & Open-Source Contributions

My roles are founder, maintainer and contributor of the below repositories solving customer problems using AWS.

https://github.com/aws-samples/aws-codepipeline-dotnet-lambda
https://github.com/aws-samples/aws-ecs-windows-aspnet
https://github.com/aws-samples/amazon-ecs-fargate-aspnetcore/
https://github.com/aws-samples/iot-dotnet-publisher-consumer
https://github.com/aws-samples/aws-iot-core-http-sigv4-dotnet-app
https://github.com/aws-samples/aws-iot-core-dotnet-app-mqtt-over-websockets-sigv4
https://github.com/aws-samples/aws-iot-core-python-node-sigv4-https
https://github.com/aws-samples/aws-iot-core-https-x509certificate-python-node